

Xiaohang (David) Zhang

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Research Interests

Statistical genomics and spatial statistics; Bayesian modeling and Network modeling for high-dimensional biomedical data, with applications to precision oncology.

Education

Boston University, Boston, MA

Jan 2026

Master of Science in Biostatistics

GPA: 3.7/4.0

- **Relevant coursework:** Probability Theory; Mathematical Statistics; Estimation Theory; Advanced Multivariate Analysis; Computational Statistics; Logistic Regression and Survival Analysis; Generalized Linear Models; Analysis of Correlated Data; Statistical Genomics; Epidemiology.

Boston University, Boston, MA

May 2024

Bachelor of Arts in Statistics; Minor in Biology & Environmental Science

GPA: 3.1/4.0

- **Relevant coursework:** Linear Algebra; Ordinary Differential Equations; Multivariate Calculus; Advanced Statistical Learning; Bayesian Statistics; Causal Inference; Stochastic Processes.

Research Experience

Boston University — Department of Biostatistics

Ongoing

Research Assistant | Advisor: Prof. Chunyu Liu | Project: *Epigenetic Aging Clocks and Alcohol Consumption*

- Investigating associations between DNA methylation-based epigenetic aging clocks and alcohol consumption, applying regression and mixed-effects models to quantify alcohol-related epigenetic age acceleration.

Boston University — Department of Biostatistics

May 2025–Present

Research Assistant | Advisor: Prof. Huimin Cheng | Project: *Community Detection for Knowledge-Graph Development*

- Refined community-detection algorithms in the Graph-RAG framework to strengthen knowledge-graph construction and support downstream biomedical applications.

Indiana University — Department of Biostatistics

May 2026–Present

Research Assistant | Advisor: Prof. Selena Wang | Project: *GNPC Hypergraph Modeling for Alzheimer's Disease Subtypes*

- Developing hypergraph-based models on Global Neurodegeneration Proteomics Consortium (GNPC) data to capture higher-order proteomic interactions and identify clinically meaningful Alzheimer's disease subtypes.

Boston University — Department of Mathematics and Statistics

May 2024–Jan 2026

Researcher | Advisor: Prof. Mark Kon | Project: *Algorithm for Detection of Breast Cancer Using Genomic Data*

- Created a gene-pathway ensemble model applying a Bayesian Dirichlet-tree framework to gene-symbol

features in plasma data for cancer-label prediction, and packaged the workflow into user-friendly desktop software.

Boston Medical Center

Aug 2025–Oct 2025

Research Analyst | Project: *Hospital Variation in Tonsillectomy and Adenoidectomy Procedures*

- Conducted chi-square and multilevel analyses to quantify how hospital region and type influence the likelihood of concurrent tube placement, adjusting for demographics and comorbidity mix.

Boston University — School of Public Health

Oct 2024–Mar 2025

Research Assistant | Advisor: Prof. Shariq Mohammed | Project: *Spatially Variable Genes Detection*

- Developed and coded a computational algorithm for detecting spatially variable genes, achieving higher detection accuracy and markedly better interpretability than the current benchmark method.

Peking University Clinical Research Institute

May 2023–Jan 2024

Intern | Projects: *Bayesian Modeling & SAP Development*

- Developed a hierarchical Bayesian model for femur-fracture outcomes using informative priors from related studies, producing stable estimates despite a small sample size.
- Authored the SAP for the Tongxinluo Phase III trial; revised a migraine trial protocol; evaluated AI lung-nodule detection via AFROC analysis.

Publications and Manuscripts

- Lin Tang, **Xiaohang Zhang**, Xuezhou Zhang, Xiaowei Yu, and Huimin Cheng. *Statistical Network Analysis for LLMs via Knowledge Graphs*. Under review (revised and resubmitted), 2025.

Conference Presentations

- Lin Tang, **Xiaohang Zhang**, Xuezhou Zhang, and Huimin Cheng. *A Human-in-the-Loop LLM Framework for Trustworthy Diabetes–Nutrition Knowledge Graph*. Accepted, Poster Track, STAI-X 2026.

Technical & Programming Skills

Python | R/RStudio | SAS | Linux | LaTeX | Algorithm development

Professional Memberships

- American Statistical Association (ASA)
- Eastern North American Region (ENAR) of the International Biometric Society